Affects Analysis: A Temporal Approach to Estimate Students' Learning

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***Abstract*— The dissimilarity between real and virtual environments is getting closer rapidly. More people are working on computers to accomplish a variety of tasks. Tasks like shopping, banking, learning, teaching, etc. are all being carried out online. The education sector is one of the sectors that has adapted to the needs and obstacles created by the impossibility to present a face to face sessions, which was necessitated because of the global pandemic. These obstacles have led to more options for online teaching and learning. Unfortunately, the underlying experiences, feelings, emotions, and moods of students during learning are not adequately captured by teachers, unlike in the classroom environment where a teacher can observe students through visual and audio cues such as eye gaze, body language, facial expressions, speech, etc. Facial emotion expressions recognition during online learning has been a subject of research in the field of machine learning, but most have not adequately considered analyzing a student's affect states with the possibility of providing appropriate feedback. This work, therefore, intends to contribute to the existing knowledge by proposing a temporal approach to estimate a student's learning affects. A model, that comprises of a video input of facial emotion expressions, feature extraction using ResNet-50 pre-trained CNN model, and a bi-directional Long Short-Term Memory classifier to detect and classify affect states of boredom, confusion, frustration, and engagement for estimating student's learning is proposed. The baseline model was trained, validated, and tested using the DAiSEE dataset and shows a test accuracy of 88% on 6546 test samples. The work further maps the affect states into positive and negative learning affects. It is anticipated that such a model will serve as feedback to the teachers and enhance the estimation of the students' learning, especially during e-learning sessions.**

Keywords*— *Affect states, Convolutional Neural Network, Bi-direction Long Short-Term Memory, Temporal data.

I. Introduction

The world system has experienced significant disruptions due to the lockdown and closure of many establishments and sectors that were caused by the global pandemic. Sectors like business firms, education systems, entertainment industries, telecommunication systems, medical sectors, and manufacturing industries to mention a few. This halt in the system has forced the world to seek solutions to get going again.

Consequently, in sectors like the education system, schools have experienced closure due to the pandemic and have given rise to learning losses and inequality to education[1]. Therefore, to forestall these long-term negative effects, as most other sectors did, teaching and learning institutions had to go online to find solutions to the long-term negative effects of the restrictions. As a result of this abruptness of the situation, teachers were forced to transit to an emergency remote teaching and learning system[2].

Unfortunately, in remote teaching and learning, otherwise known as online learning, a major limitation is the absence of personal interaction between the teachers and the students. Teachers' interactions and observations of students during teaching and learning sessions are one of the major tools used to assess the effects of the student's learning. Students' learning assessments are an important part of education all over the world as stressed by authors in [3]–[5].

For a teacher to teach effectively and be able to meet the students' needs during a teaching and learning session, there is the need for the teacher to be able to quickly access the state of their audience [6], [7]. Teachers assess students' understanding in a classroom through their facial emotion expressions, their body movements, eye gaze, and several other methods, but in online learning, this is very difficult. Several factors are required in assessing student's achievement in online learning, according to authors in [3],[8]. It was also emphasized that mood has a significant impact on learning, and this can be determined from facial emotion expressions. Getting a good understanding of students' affect during learning continues to be an important objective to teachers. Students experience different emotions either cognitive or affective when they are learning.

The absence of this ability to adequately estimate the student affects has resulted in diverse research in machine learning and computer vision. Authors in [9], [10] suggested that facial emotion expressions give important information for teachers to discover the learning conditions or states of students. In [11], the authors proposed a method that locates the facial expressions of students to recognize their comprehension state using the facial expressions in static images of human faces. Their goal was to group the facial expressions states to comprehensible and incomprehensible. They concluded that, aside from words, the facial expression is the most important source of information, and used this output as feedback to the learning management system. Therefore, this study considers the use of facial emotion expressions for the analysis of the learning affects of students.

The work in [12] itemized different emotions that were being displayed by the students through their facial expressions as majorly; frustration, confusion, delight, boredom, curiosity, engagement, anxiety, and surprise. These emotions are not taken cognizance of during online learning. Research work in computer vision has attempted to recognize emotions, especially during learning but most have not estimated the learning affects from these emotion

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outcomes as a way of accessing and evaluating the learning outcomes.

This work, therefore, proposes a solution to the shortfalls in the assessments of students' comprehension during online classes. This work proposes a model to perform affect analysis to estimate students' learning which is applicable in online learning, using a temporal approach. The work proposes a CNN-Bi-LSTM model to analyze the learning affects of students and considers mapping of the estimated affect states to positive and negative.

The work trained, validated, and tested the temporal model with dataset of labeled affect emotions (DAiSEE) of boredom, engagement, confusion, and frustration[13]. The study further tested the trained model with DISFA+ facial emotion expression data (Denver Intensity of Spontaneous Facial Action) which comprises of the basic emotions and discovers that the affect emotions can also be mapped effectively with the seven basic emotions of fear, anger, disgust, sadness, happiness, surprise, and neutral emotions.

The remaining part of the paper is organized as follows: Section 2 describes some of the previous works carried out on facial emotion analysis for use in online learning. The proposed approach is outlined in Section 3, while the details of the experiments are presented in Section 4. Section 5 provides the discussion of results and Section 6 provides a conclusion of the study

II. REVIEW OF RELATED WORKS

Affects are described in [14], [15] as the attitudes, emotions, and values present in a learning domain. In online learning, methods used to evaluate students' comprehension include quizzes, assignments, tests, questionnaires, biophysical sensors, and examinations to determine their learning affects. Unfortunately, the real-time evaluation of students, through their facial emotional expressions has been a difficult task, especially in the field of machine learning. Over the last decade, researchers have attempted to solve the problem of facial emotion expressions recognition in a learning environment using different approaches. Invariably, affect analysis of students in e-learning, mapping of emotions to learning affects and feedback to assist teachers is an area that has received little to no work.

Authors in [16] proposed an information system to recognize the emotional states of students. The system comprises of information about the instant and weighted emotional states that were based on the facial expressions of students. The work employed K-Nearest Neighbor (KNN) and Support Vector Machine (SVM) for their study and produced an accuracy of 96.38% and 98.24% respectively. Authors in [17] proposed a system that identifies the facial emotion expressions of students in real-time using a SVM. Their model achieved an accuracy of 97% and 98% for facial emotion expressions and head gesture classifications, respectively. Authors in [18] focused on the emotion detection of learners with the use of biophysical signals. The biophysical signal considers the emotional changes during learning. The work used neural network classification to classify learners' emotions and to predict their learning styles. Their work intended to show the effectiveness of facial emotion recognition for identifying the affective level of a student using SVM and presented an accuracy in terms of average gain for their sample as 1.3693. The average gain was used to determine the course delivery strategy based on the learner's learning style.

The learners' behaviors and emotional expressions throughout online learning were captured in the work of authors in [19]. The research looked at combining CNN with fuzzy systems. The learner's facial expression was detected using the CNN, and the fuzzy system was utilized to predict the next learning level using the extracted facial expression states from the CNN.

Work in [20] employed an emotion detection module applicable to e-learning. CNN was used to detect emotion in an e-learning setting and their model achieved an accuracy of 84.55%. Work in [21] assessed the emotions of learners and identified the emotional changes that occur during online learning. The work classified basic emotions using the CNN model and then identified the state of mind of the learners. The model achieved an accuracy of 65% and 62% for the emotion classification and state of mind identification respectively.

In [22], the authors proposed using a temporal relational network to identify students' facial emotion changes during an e-learning session to measure the students' learning affects and engagement level. The study considered the use of a single scaled and multi-scale temporal relation network and compared it with a Multi-layer Perceptron (MLP) baseline model. DISFA dataset was used for the experiment. The experiment achieved an accuracy of 92.7%, 89.4%, and 86.6% on the multi-scale TRN, single-scale TRN, and MLP respectively.

Authors in [23] examined the estimation of students' learning affect using facial emotions expressions in an e-learning session. The study trained and tested a CNN model on the FER2013 dataset of basic emotions and achieved an accuracy of 64% on the test data.

The majority of previous facial emotion recognition research has concentrated on the basic emotions of anger, fear, sadness, happiness, disgust, surprise, and neutral emotions. [24]. However, some studies show that these basic emotions are not the most frequent during a short learning session, especially during an online learning session. Rather than these basic emotions, affective states of happiness, engagement, boredom, anxiety, confusion, frustration, and curiosity were identified to be more prevalent during online learning [25].

Authors in [26] opined that they were not certain of the accuracy of the detection of learning-centered affective states in online learning as compared with the detection of the basic emotional expressions which have been investigated for over ten years [26]. However, authors in [27] looked at the estimation of engagement of students within the online environment by using facial images. The work measured the state of engagement of learners during online learning, using the InceptionNet architecture and achieved accuracies of 36.5%, 47.1%, 70.3%, and 78.3% of engagement levels for boredom, engagement, confusion, and frustration emotions.

The use of a deep learning technique on FaceNet embeddings and facial landmark points for academic emotion detection using the DAiSEE dataset was discussed in [28]. The dataset was labeled with emotional states of engagement, boredom, frustration, and confusion. The work was able to detect instances of boredom and frustration which learners experienced in an online class and achieved an accuracy of 52.1%, 70.67% on both emotional expressions, respectively.

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Most previous research and models developed have not made much progress in assessing students' facial emotional expressions to understand their affects in terms of positive or negative emotions. Furthermore, most of the existing models lack feedback to teachers in online settings.

Therefore, in the proposed work, the authors intend to contribute to existing works by using a temporal approach for the analysis of students' learning affects from classified emotions (boredom, engagement, confusion, and frustration). Furthermore, its application to online learning for the estimation of students' learning outcomes is explored, which is envisaged to provide feedback to teachers and help cause an improvement to the platform.

III. METHODS OF APPROACH

The proposed research is composed of the following stages:
Stage 1: feature extraction using pre-trained ResNet- 50, Bi-LSTM classifier for affect emotion expressions recognition and classification, mapping of classified emotions into positive and negative learning affects. Stage 2: testing of the model with test datasets that were not been part of the training set. Stage 3: testing the model with a new dataset sample. The DISFA+ dataset of the basic emotions was used for the second testing. Figure 1 shows the framework of the proposed model.

A. Dataset

The dataset used for this work is DAiSEE [13]. DAiSEE is a publicly available dataset and it comprises 9,068 video sequences, which were recorded from 112 subjects from Asia. The labeling of the dataset was done through crowd annotations and was labeled with the four learning-related affective states of boredom, engagement, confusion, and frustration[13]. The dataset is freely available for research purposes. The work chooses this dataset for the baseline training, evaluation and initial testing because this dataset has been developed for user engagement recognition. Using state-of-the-art video classification, the author in [13] established a benchmark result on this dataset.

Authors in [29] used DAiSEE to test their model and achieved a four-class classification accuracy of 58.84% and a mean square error (MSE) of 0.0422. On DAiSEE, they reported that their work exceeds several other works in terms of engagement prediction.

In [30], the authors described ResNet + TCN, a new end-to-end Spatio-temporal hybrid architecture for assessing student participation in an online classroom scenario. The model was tested on the DAiSEE dataset and reported improved metrics than the current state-of-the-art engagement classifications.

This study focuses on the training of a temporal model for the detection and classification of these affective states, which were identified in learning and are consequently mapped into positive and negative affects.

B. Pre-Processing

The DAiSEE dataset contains video sequences which are of a resolution of 640 x 480 pixels and a duration of 10 seconds each and the frames were extracted at 10fps. The total number of frames is 2,723,882 from all the videos. The study selected a total of 32,730 frames for the experiment.

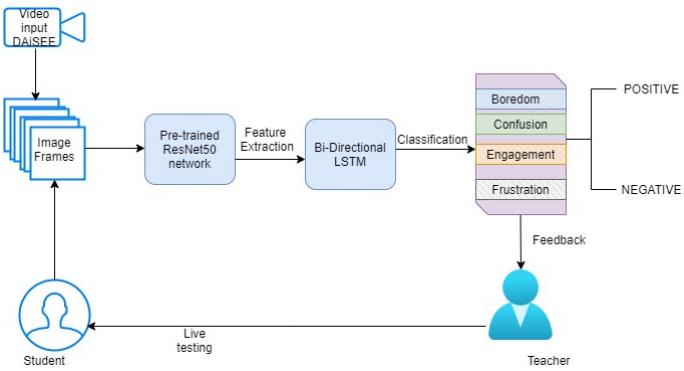

Figure 1: The proposed Temporal framework

C. Feature Extraction

The facial image of 24,329 samples which are of size 480 x640x3 are used for the training and validation and are reshaped to size 240x 320 x 3 and passed through the ResNet-50 network for the extraction of discriminate features. ResNet-50 is a network with the main idea of skipping blocks of convolution layers by using shortcut connections [31]. ResNet-50 was trained on the ImageNet database and can classify1000-categories of objects [31]. The network produced a feature size of 24329 x 2048.

D. Bi-Direction Long Short-term Memory

The extracted features are passed into the Bi-LSTM network in which the classification takes place. The bi-directional LSTM is an expansion of the LSTM model which is made up of two LSTMs. The first LSTM is applied to the input sequence data in the forward direction, and in the second LSTM takes in the reverse structure of the input sequence. This form of LSTM helps to improve learning long-term dependencies and can improve the accuracy of a model [32]. The forward hidden layer h_t^f processes the input in ascending order, that is t = 1,2,3….T, while the backward hidden layer h_t^b process the input sequence in descending order that is, t = T,…..3,2,1. Lastly, both are combined to generate output Y_t. A Bi-LSTM is implemented using the following equations:

$$h_t^f = \tanh\left(W_{xh}^f xt + W_{hh}^f h_{t-1}^f + b_h^f\right. \quad (1)$$

$$h_t^b = tanh\left(W_{xh}^b\, xt + W_h^b hh_{t+1}^b + b_h^b\right) \quad (2)$$

$$y_t = W_{hy}^f h_t^f + W_{hy}^b\, h_t^b + by \quad (3)$$

E. Evaluation Metrics

The work considered the use of accuracy, precision, recall, and F-1 score to evaluate the proposed model. The computations for precision (P), recall (R) and F-1 score are given in the equations below.

$$\text{Precision(P)} = \frac{TP}{TP+FP} \quad (4)$$

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$$\text{Recall(R)} = \frac{TP}{TP+FN} \quad (5)$$

$$\text{F-1 score} = \frac{2*P*R}{P+R} \quad (6)$$

True Positive, True Negative, False Positive, and False Negative are all abbreviated as TP, TN, FP, and FN, respectively.

IV. DETAILS OF EXPERIMENT

A CPU Intel Core i7, 8th generation, with 16G RAM and Linux operating system was used to carry out the implementation for this work. A tensor-flow environment was used in running the codes, with a tensor-board to visualize the results during training and system evaluation. The work stages include:

A. Training and Validation

The extracted features of size 24329 x 2048 were split into training data size of 19638 x 2048 and validation data size of 6546 x 2048 which are passed to the Bi-LSTM network, for the classification. The loss function was identified using a batch size of 128, maximum epochs of 100, and binary cross-entropy, sigmoid function was utilized to compute the probability of the output layer. Early stopping was used to avoid overfitting. The study conducted five-fold cross-validation on the samples to avoid any randomness bias during the training and validation tests. The performance metrics that were used for the learning accuracy are F-1 score and confusion matrix. Both the temporal and spatial features of the samples were considered for training and validation by the CNN- Bi-LSTM. The model was then tested using the remaining part of the dataset that has not been exposed to the network before and the accuracy was recorded.

V. DISCUSSION OF THE RESULTS

The results of the tests conducted on the optimized model are discussed in this section.

A. Testing

The model was tested with 6546 unseen samples of the DAiSEE dataset. The CNN-Bi-LSTM model achieved an 88% accuracy on the test data. The confusion matrix for the result is presented in Figure 2 and the summary of the test results is given in Table 1.

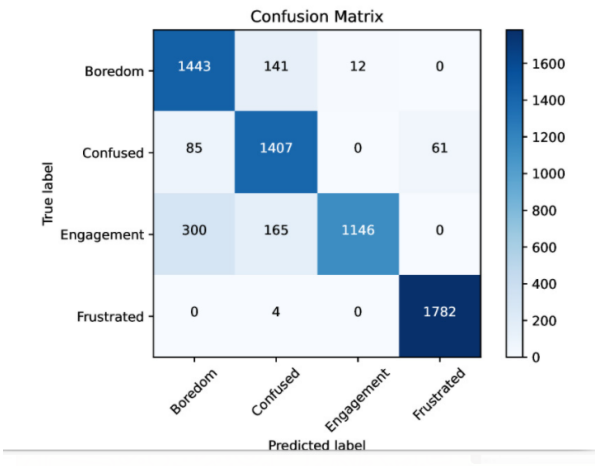

Figure 2: Confusion Matrix for the test result

TABLE 1. SUMMARY OF THE TEST RESULTS

Emotions	Training Results			
	Precision	***Recall***	***F1-score***	***Number of samples***
Boredom	0.79	0.90	0.84	1596
Confusion	0.82	0.91	0.86	1553
Frustration	0.99	0.71	0.83	1611
Engagement	0.97	1.0	0.98	1786
Accuracy			0.88	Total=6546

B. Mapping of Emotions to Learning Affects

The detected emotions were mapped into positive and negative learning affects. This mapping is initially formulated based on literature and is will be revised based on extensive experiment. This is aimed to serve as high-level feedback to teachers to determine the overall acceptance and quality of the lecture materials and the methods of the approach used.

Authors in [33] opined that students that experience high levels of boredom, result in having a high risk of negative consequences, some of which include low grades, absenteeism, and eventually, dropping out of school. This, they conclude as a negative learning affect. Authors in [25] also stated that boredom is experienced by students when the material does not interest them, or when they cannot cope with any task involved in the learning process, which eventually leads to negative learning affects. Difficulty in comprehending the material often leads to confusion. When students come across a challenge in the course of their learning activities or are unsure about the content being disseminated according to [34], [35]. Thus, they concluded that it is a negative learning outcome.

Authors in [25], [36] stated that in situations when students make mistakes repeatedly, frustration sets in. Sometimes, when the students get confused, especially in understanding or comprehending a lecture, this results in frustration. Therefore, these authors in [25],[36]concluded that frustration corresponds to a negative learning affect. Authors in [37]–[39] iterated in their work that there is a flow- like state of intense engagement, especially when there are clear learning outcomes and a good understanding of the lecture content.

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TABLE 2: MAPPING EMOTIONS TO LEARNING AFFECT

	Authors				
	Current study	***[24][31]***	***[32][33]***	***[24][34]***	***[35] [36][37]***
Boredom	Negative	Negative	-	Negative	-
Confusion	Negative	Negative	Negative	Negative	-
Frustration	Negative	Negative	-	Negative	-
Engagement	Positive	Positve	-	Positive	Positive

This they categorized as being a positive learning outcome. Table 2 shows the summary of mapping based on the above works.

Drawing from various studies, the current study, therefore, re-iterates that, varying affective states can arise when students are learning. Furthermore, the study suggests that affects and learning can be interwoven.

C. Further Testing with Basic Emotions

The study further explores the mapping between emotions and affective states. The work tested the trained temporal model using an unlabeled DISFA+ dataset which comprises samples of surprise, anger, disgust, fear, happiness, sadness, and neutral emotions. The work experimented on 5847 unlabeled samples of the DISFA+ facial emotion expressions dataset. The model maps samples of happy and surprise emotions primarily as engagement affective states as a result of the experiment. Happiness and surprised emotions, as well as engagement affective states, were regarded positive learning affects, which aligns with our original mapping.

Fear and sadness emotional states were mapped to confusion affective state. Confusion affective state was considered negative learning affects, as well as fear and sadness emotions, which reconciled our initial mapping. Disgust emotion was mapped to boredom affective state, and boredom affective state was regarded to be a negative learning affect, hence disgust emotions, which also agrees with our initial mapping. Anger was mapped to frustration affective state. Frustration was considered a negative learning affects, so also anger emotion, this also aligns with our initial mappings.

TABLE 3: RESULT OF THE EXPERIMENT ON TEST DATA

Emotions (DISFA+)	**Classification by Temporal model**				
	Number of test samples	***Boredom***	***Confusion***	***Engagem -ent***	***Frustration***
Anger	1403	-	-	-	**1403**
Disgust	782	**782**	-	-	-
Fear	758	-	**758**	-	-
Happy	1362	-	-	**1362**	-
Sadness	751	-	**751**	-	-
Surprise	791	-	-	**791**	-

The study, therefore, opines that the basic emotions are also embedded in the learning emotions that were being analyzed in this study. Table 3 summarizes the result of the experiments.

D. Feedback

The detected emotions were mapped into positive or negative learning affects based on the experiments conducted with the help of the temporal model. Teachers will be able to analyze the overall learning affect of a cohort using this tool, which will aid them in making major discissions or adjustments regarding their teaching methods and teaching materials.

E. Real-time Classes Working Description

Based on the proposed model in this work, for the real-time classes, the model could be adopted in an online classroom setting with computer systems and a webcam to capture the facial expressions of students and internet connection. During an online class session, the student sits in front of the computer system to learn. The webcam is activated as soon as the video starts to collect the learner's facial images. The model captures the facial images of the student progressively in real-time through the webcam. The facial videos are split into frames of images, and reshaped, then passed to the pre-trained CNN, for the features extraction. The extracted features are further sent to the Bi-LSTM model for classification of the facial features to various affects emotions. The displays show changes in the affective states of the student as the lecture progresses. At the end of the lecture, the model displays the maximum occurring affects emotion to the teacher and the analysis of the learning affects is recorded based on the mappings provided by the research. This is expected to provide an understanding of the learning affects of the students during the lecture.

VI. CONCLUSION AND FUTURE WORKS

The study used machine learning models, which include the DAiSEE dataset, pre-trained CNN and Bi-LSTM for classification, and mapping of classified emotions into learning affects. The study achieved a classification accuracy of 88% on the test dataset. The work also tested the temporal model with the DISFA+ dataset to test the basic emotions and compared it with the affective states emotions . Most research being carried out in facial emotion expressions recognition in the education domain has not adequately analyzed learning affect using transfer learning and Bi-LSTM classifier, and most have focused on the basic emotions. Here the study delved into the learning emotions such as engagement, boredom, confusion, and frustration. The study analyzed the changes of the facial emotion expressions exhibited by students in online learning, using a temporal model. The input comprising of video of facial emotion expressions, a pre-trained CNN, a Bi-LSTM classifier and mapping of classified emotions to positive or negative learning affects. It is anticipated that this work will improve the online

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platform, and hopefully, in the future, it can be improved upon especially since more features such as reasoning can be included in the model.

REFERENCES

[1] Z. Blask, P. da Costa, and S. V. Schnepf, "Learning loss and educational inequalities in Europe: Mapping the potential consequences of the COVID-19 crisis," 2021.

[2] A. E. Al Lily, A. F. Ismail, F. M. Abunasser, and R. H. A. Alqahtani, "Distance education as a response to pandemics: Coronavirus and Arab culture," *Technol. Soc.*, vol. 63, p. 101317, 2020.

[3] P. Kaur, H. Kumar, and S. Kaushal, "Affective state and learning environment-based analysis of students' performance in online assessment," *Int. J. Cogn. Comput. Eng.*, vol. 2, pp. 12–20, 2021.

[4] S. Deshpande and A. Shesh, "Performance assessment and remedies using blended learning for professional students," in *Computing in Engineering and Technology*, Springer, 2020, pp. 783–790.

[5] M. A. Schlenz, A. Schmidt, B. Wöstmann, N. Krämer, and N. Schulz-Weidner, "Students' and lecturers' perspective on the implementation of online learning in dental education due to SARS-CoV-2 (COVID-19): a cross-sectional study," *BMC Med. Educ.*, vol. 20, no. 1, pp. 1–7, 2020.

[6] R. Klein and T. Celik, "The Wits Intelligent Teaching System: Detecting student engagement during lectures using convolutional neural networks," in *2017 IEEE international conference on image processing (ICIP)*, 2017, pp. 2856–2860.

[7] R. E. Ferdig, E. Baumgartner, R. Hartshorne, R. Kaplan-Rakowski, and C. Mouza, *Teaching, technology, and teacher education during the COVID-19 pandemic: Stories from the field*. Association for the Advancement of Computing in Education, 2020.

[8] D. Surani and H. Hamidah, "Students Perceptions in Online Class Learning During the Covid-19 Pandemic," *Int. J. Adv. Sci. Educ. Relig.*, vol. 3, no. 3, pp. 83–95, 2020.

[9] Z. Yang, J. Yu, C. Yang, Z. Qin, and Y. Hu, "Multi-modal Learning with Prior Visual Relation Reasoning," p. 15.

[10] D. Mitchell, *What really works in special and inclusive education: Using evidence-based teaching strategies*. Routledge, 2014.

[11] M. Sathik and S. G. Jonathan, "Effect of facial expressions on student's comprehension recognition in virtual educational environments," *SpringerPlus*, vol. 2, no. 1, pp. 1–9, 2013.

[12] A. C. Graesser and S. D'Mello, "Emotions during the learning of difficult material," in *Psychology of learning and motivation*, vol. 57, Elsevier, 2012, pp. 183–225.

[13] A. Gupta, A. D'Cunha, K. Awasthi, and V. Balasubramanian, "Daisee: Towards user engagement recognition in the wild," *ArXiv Prepr. ArXiv160901885*, 2016.

[14] W. J. Popham, "Assessing Student Affect.," *Educ. Leadersh.*, vol. 66, no. 8, pp. 85–86, 2009.

[15] J. A. Pooley and M. O'Connor, "Environmental education and attitudes: Emotions and beliefs are what is needed," *Environ. Behav.*, vol. 32, no. 5, pp. 711–723, 2000.

[16] U. Ayvaz, H. Gürüler, and M. O. Devrim, "Use of facial emotion recognition in e-learning systems," *Інформаційні Технології І Засоби Навчання*, no. 60, вип. 4, pp. 95–104, 2017.

[17] S. P. Deshmukh, M. S. Patwardhan, and A. R. Mahajan, "Feedback based real-time facial and head gesture recognition for e-learning system," in *Proceedings of the ACM India Joint International Conference on Data Science and Management of Data*, 2018, pp. 360–363.

[18] A. Ray and A. Chakrabarti, "Design and implementation of affective e-learning strategy based on facial emotion recognition," in *Proceedings of the International Conference on Information Systems Design and Intelligent Applications 2012 (INDIA 2012) held in Visakhapatnam, India, January 2012*, 2012, pp. 613–622.

[19] M. Megahed and A. Mohammed, "Modeling adaptive E-learning environment using facial expressions and fuzzy logic," *Expert Syst. Appl.*, vol. 157, p. 113460, 2020.

[20] A. Sun, Y.-J. Li, Y.-M. Huang, and Q. Li, "Using facial expression to detect emotion in an e-learning system: A deep learning method," in *International Symposium on Emerging Technologies for Education*, 2017, pp. 446–455.

[21] M. Mukhopadhyay, S. Pal, A. Nayyar, P. K. D. Pramanik, N. Dasgupta, and P. Choudhury, "Facial Emotion Detection to Assess Learner's State of Mind in an Online Learning System," in *Proceedings of the 2020 5th International Conference on Intelligent Information Technology*, 2020, pp. 107–115.

[22] A. Pise, H. Vadapalli, and I. Sanders, "Facial emotion recognition using temporal relational network: an application to E-learning," *Multimed. Tools Appl.*, pp. 1–21, 2020.

[23] B. E. Zakka and H. Vadapalli, "Estimating Student Learning Affect Using Facial Emotions," in *2020 2nd International Multidisciplinary Information Technology and Engineering Conference (IMITEC)*, 2020, pp. 1–6.

[24] P. Ekman and W. V. Friesen, *Unmasking the face: A guide to recognizing emotions from facial clues*, vol. 10. Ishk, 2003.

[25] S. D'Mello, "A selective meta-analysis on the relative incidence of discrete affective states during learning with technology.," *J. Educ. Psychol.*, vol. 105, no. 4, p. 1082, 2013.

[26] R. Reisenzein, M. Studtmann, and G. Horstmann, "Coherence between emotion and facial expression: Evidence from laboratory experiments," *Emot. Rev.*, vol. 5, no. 1, pp. 16–23, 2013.

Authorized licensed use limited to: University of Witwatersrand. Downloaded on May 16,2023 at 17:10:30 UTC from IEEE Xplore. Restrictions apply.

[27] M. A. A. Dewan, M. Murshed, and F. Lin, "Engagement detection in online learning: a review," *Smart Learn. Environ.*, vol. 6, no. 1, pp. 1–20, 2019.

[28] F. H. Leong, "Deep learning of facial embeddings and facial landmark points for the detection of academic emotions," in *Proceedings of the 5th International Conference on Information and Education Innovations*, 2020, pp. 111–116.

[29] J. Liao, Y. Liang, and J. Pan, "Deep facial spatiotemporal network for engagement prediction in online learning," *Appl. Intell.*, pp. 1–13, 2021.

[30] A. Abedi and S. S. Khan, "Improving state-of-the-art in Detecting Student Engagement with Resnet and TCN Hybrid Network," *ArXiv Prepr. ArXiv210410122*, 2021.

[31] J. Peng *et al.*, "Residual convolutional neural network for predicting response of transarterial chemoembolization in hepatocellular carcinoma from CT imaging," *Eur. Radiol.*, vol. 30, no. 1, pp. 413–424, 2020.

[32] M. Rahman, Y. Watanobe, K. Nakamura, and others, "A Bidirectional LSTM Language Model for Code Evaluation and Repair," *Symmetry*, vol. 13, no. 2, p. 247, 2021.

[33] E. C. Daschmann, T. Goetz, and R. H. Stupnisky, "Testing the predictors of boredom at school: Development and validation of the precursors to boredom scales," *Br. J. Educ. Psychol.*, vol. 81, no. 3, pp. 421–440, 2011.

[34] S. K. D'MELLO and A. C. Graesser, "Confusion," in *International handbook of emotions in education*, Routledge, 2014, pp. 299–320.

[35] S. D'Mello, B. Lehman, R. Pekrun, and A. Graesser, "Confusion can be beneficial for learning," *Learn. Instr.*, vol. 29, pp. 153–170, 2014.

[36] N. L. Stein and L. J. Levine, "Making sense out of emotion: The representation and use of goal-structured knowledge," in *Psychological and biological approaches to emotion*, Psychology Press, 2013, pp. 63–92.

[37] J. Nakamura and M. Csikszentmihalyi, "The concept of flow," in *Flow and the foundations of positive psychology*, Springer, 2014, pp. 239–263.

[38] M. Csikszentmihalyi, "Flow and education," *NAMTA J.*, vol. 22, no. 2, pp. 2–35, 1997.

[39] M. Csikszentmihalyi, "The contribution of flow to positive psychology.," 2000.

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